Github: https://github.com/ReneStander/BMG_R_Python_2026

Introduction to Python Programming

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Outline

1. Introduction to Python
2. Basics of the programming language
3. Data manipulation
4. Data visualization
5. Basic statistical analysis

Program

09:00 – 10:30	Session 1
10:30 – 10:45	Break
10:45 – 12:00	Session 2
12:00 – 13:00	Lunch
13:00 – 14:30	Session 3
14:30 – 14:45	Break
14:45 – 16:00	Session 4

Resources

- Downey, A.B., 2012. *Think python*. O'Reilly Media, Inc.
  https://allendowney.github.io/ThinkPython/#

- Adhikari, A., DeNero, J. and Wagner, D., 2022. *Computational and Inferential Thinking: The Foundations of Data Science*, Second edition, University of California, Berkeley.
  https://inferentialthinking.com/

Introduction to Python

Why Python?

- Mature programming language.
- Has excellent properties for newbie programmers.
- Currently one of the most flexible programming languages.
- Has a large ecosystem of libraries making it easy for Data Scientists.

Applications developed with Python you may know…

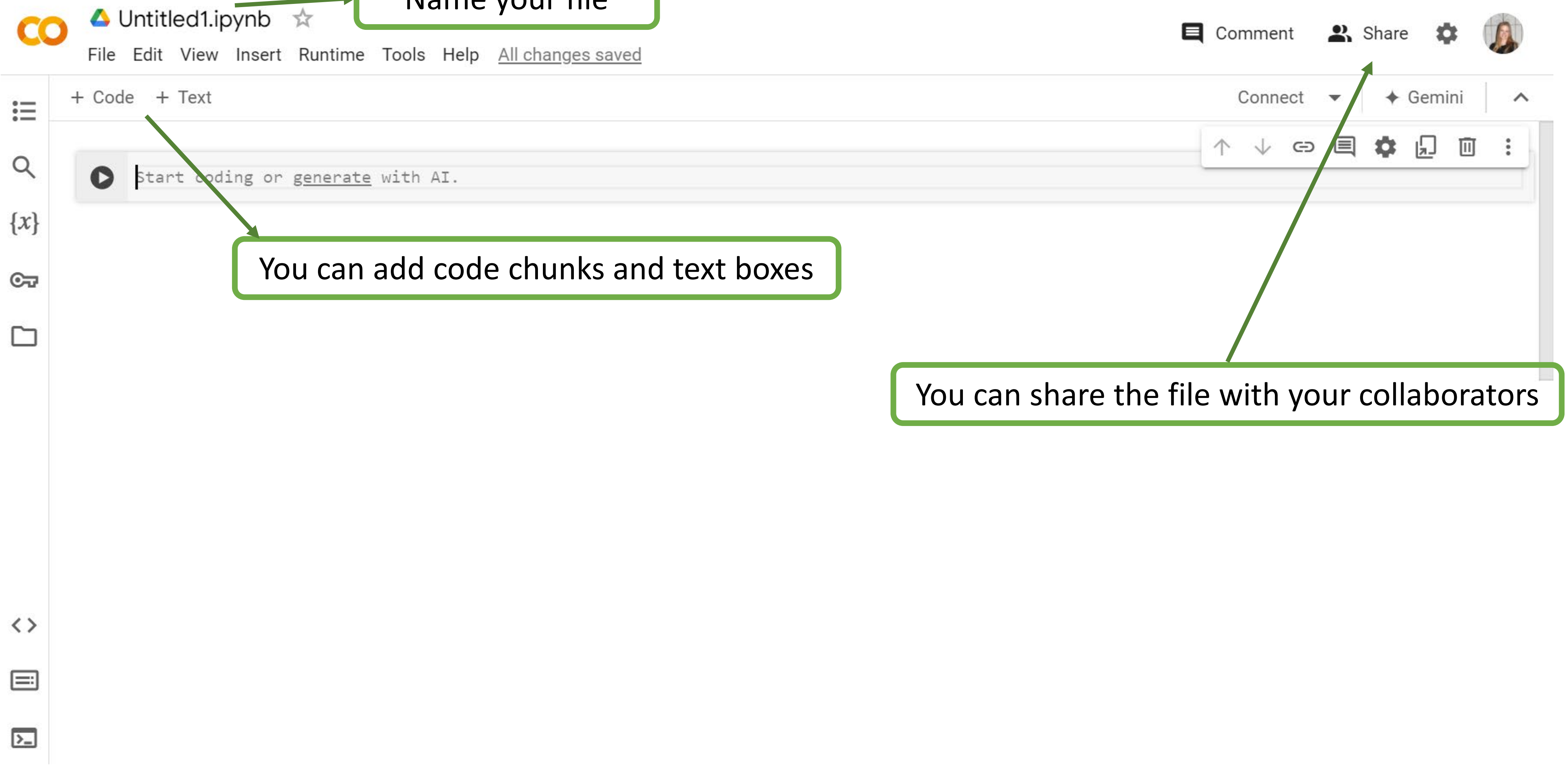
Running Python in Google Colab

Google Colab

- Hosted **Jupyter Notebook** service.
- Required **no** hardware or software to be installed on your device.
- Provides free access to computing resources such as GPUs.

https://colab.google/notebooks/

Running Python in Google Colab

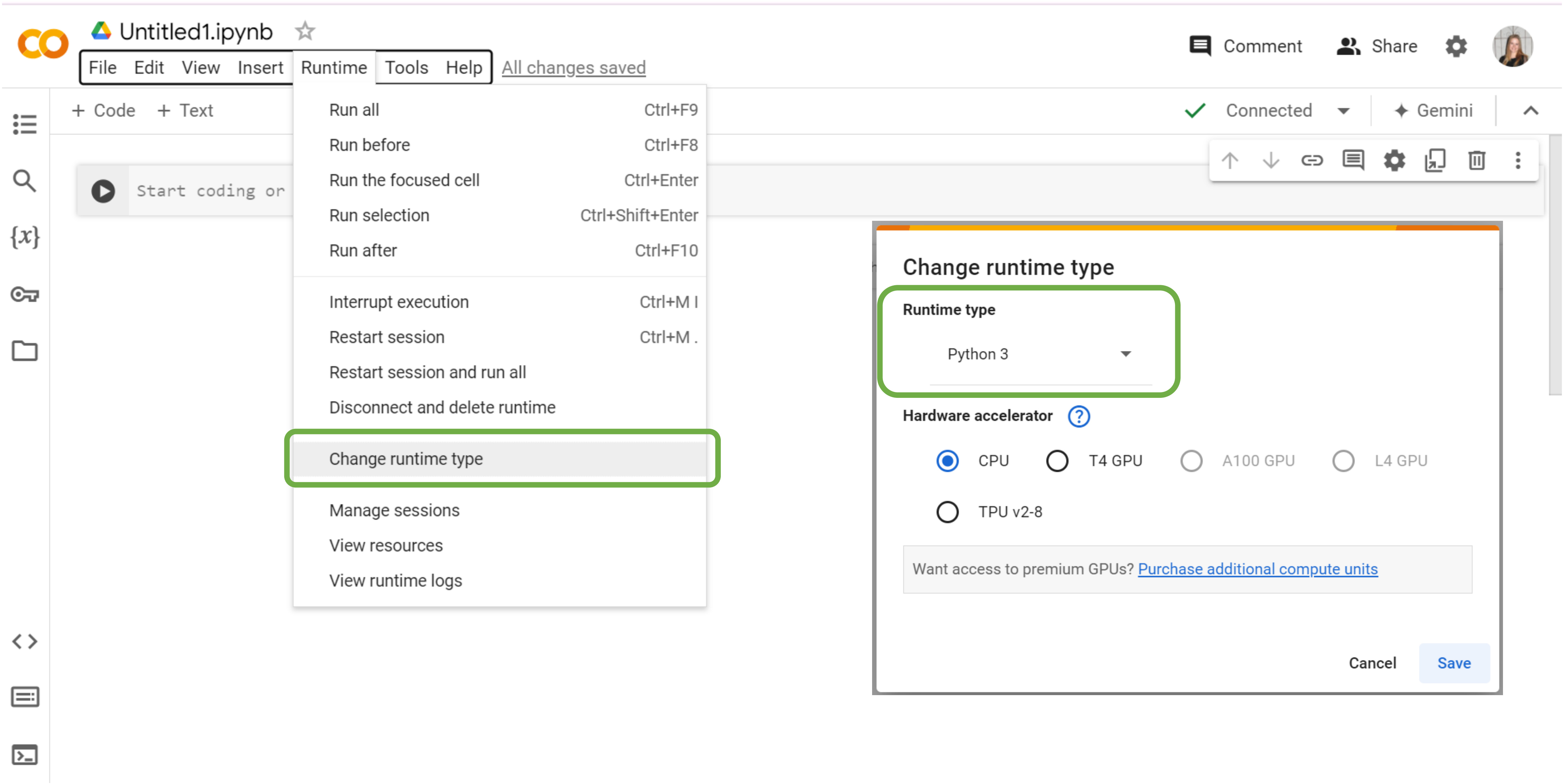


Name your file

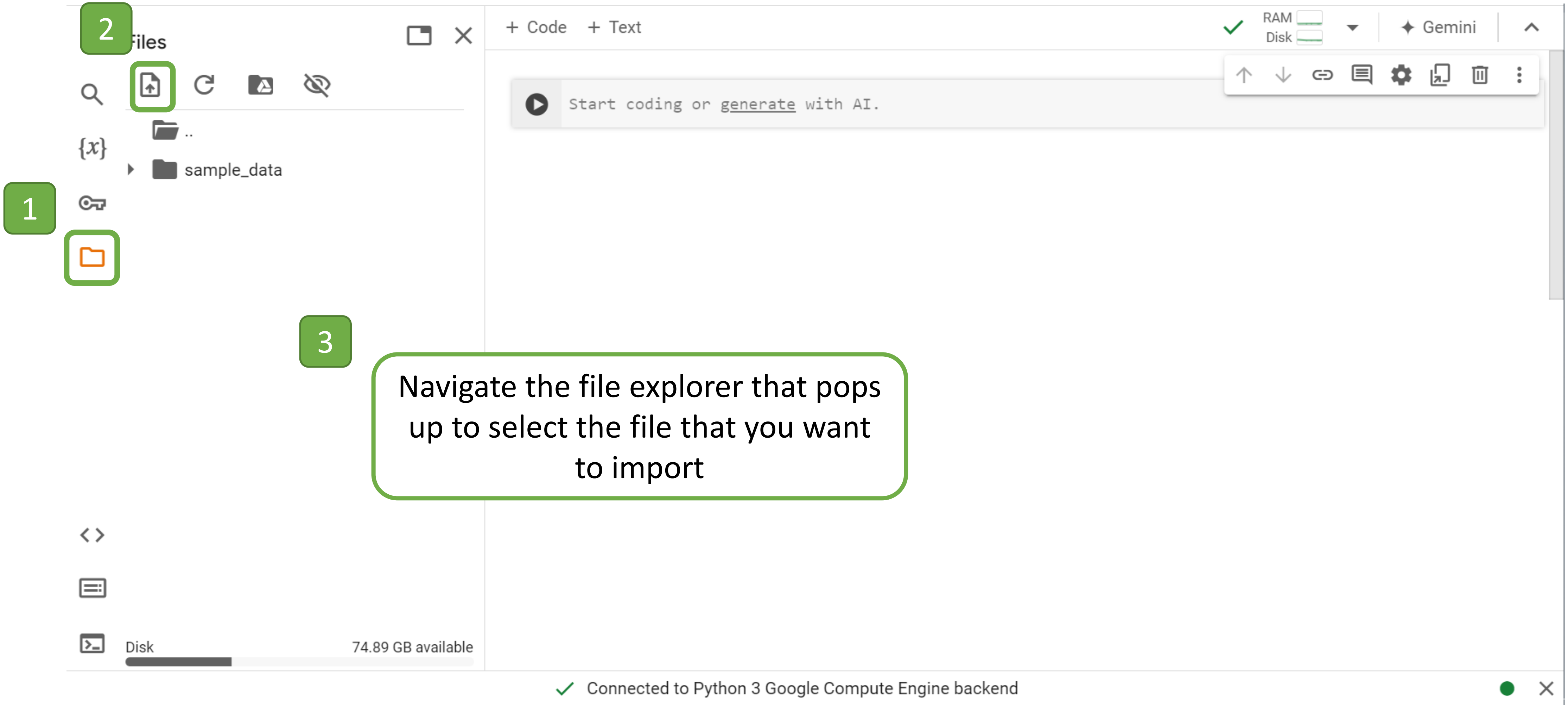
You can add code chunks and text boxes

You can share the file with your collaborators

Running Python in Google Colab



Running Python in Google Colab



Navigate the file explorer that pops up to select the file that you want to import

Installing Python

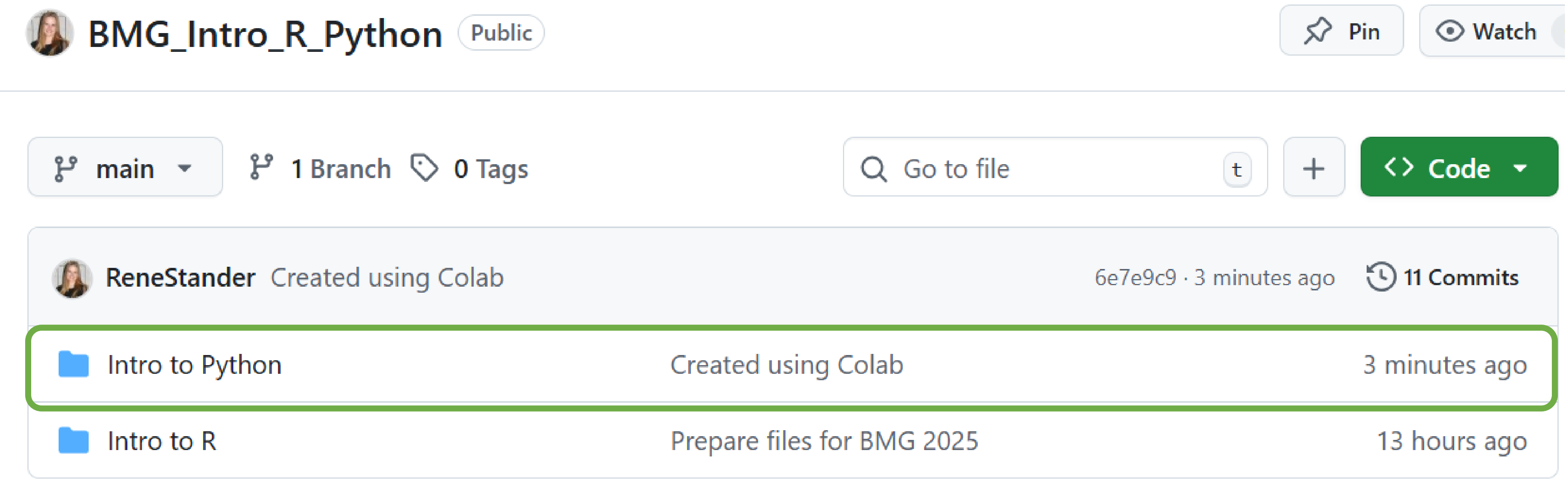
1. Download Anaconda navigator. https://www.anaconda.com/download

2. Access the Python IDE, **Spyder**, through the Anaconda navigator.

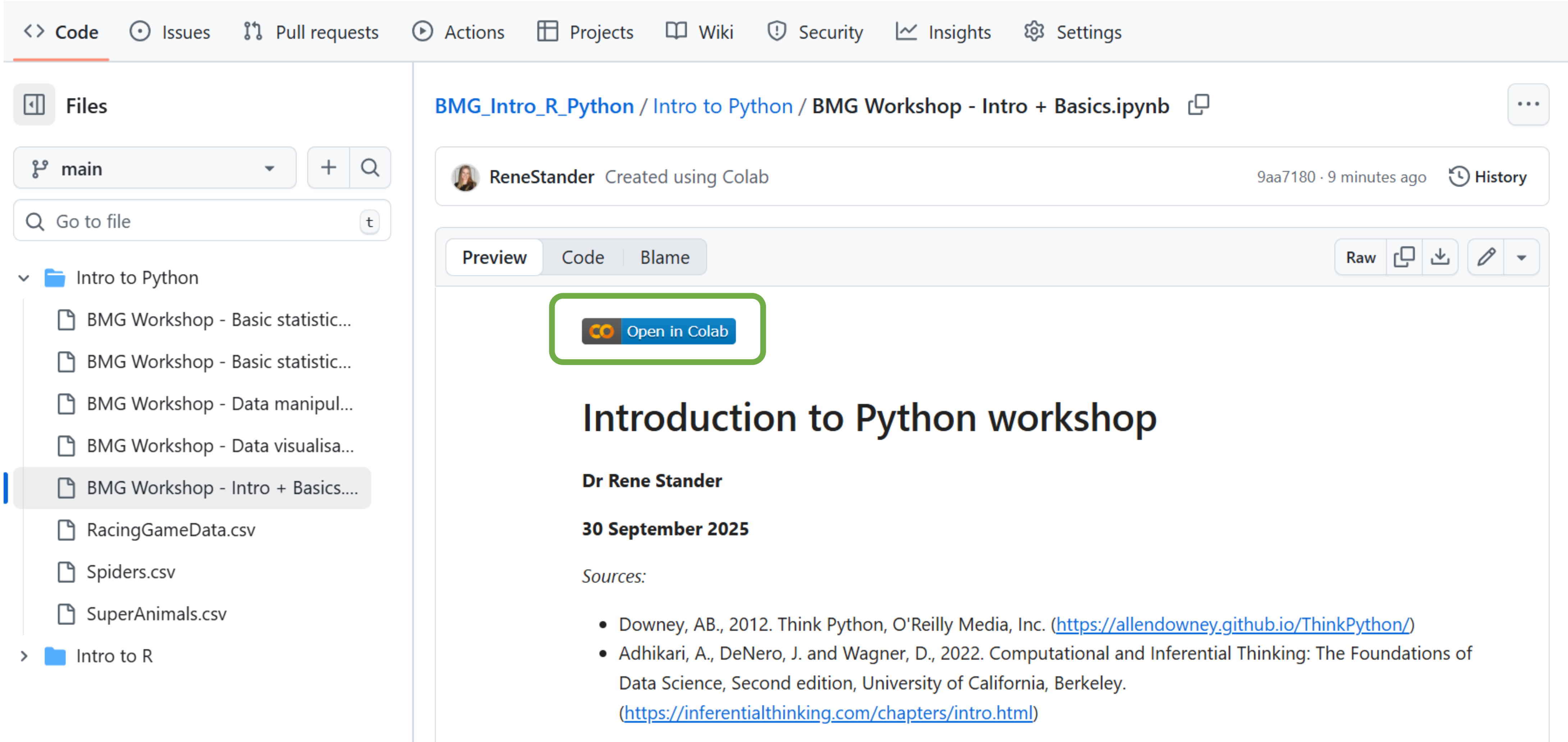
Fundamental Python Libraries for Data Scientists

- **NumPy:** Provides support for multidimensional **arrays with basic operations on them** and useful linear algebra functions.
- **SciPy:** Provides a collection of **numerical algorithms** and domain-specific toolboxes, including signal processing, optimization and statistics.
- **Matplotlib:** Enables **data visualization**.
- **Pandas:** Provides high performance **data structures** and data analysis tools.
- **Scikit-Learn:** Offers simple an efficient tools for **common tasks in data analysis** such as classification, regression, clustering, and many more…

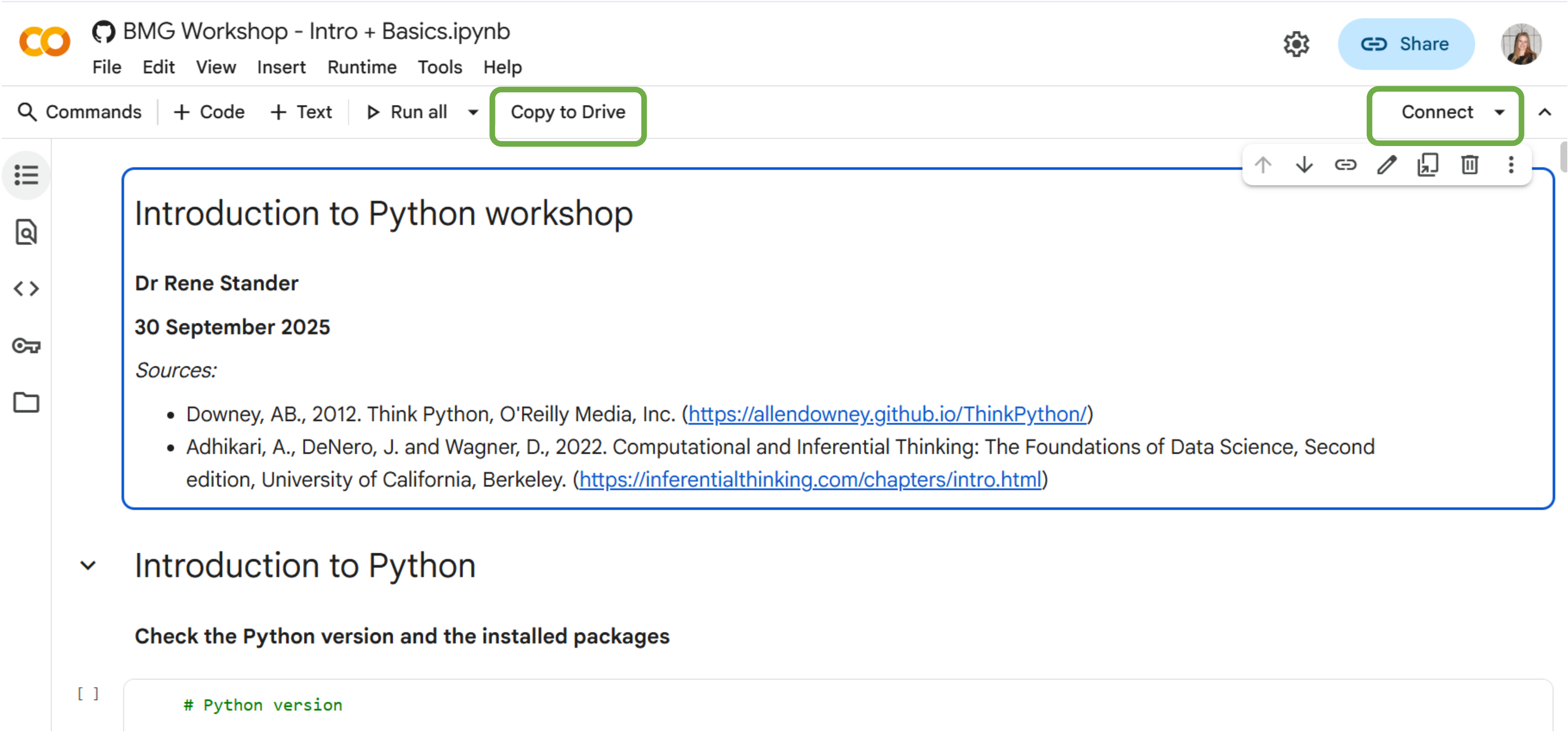
Coding scripts for this course



Coding scripts for this course



Coding scripts for this course



Intro + Basics

Open the notebook named **`BMG Workshop – Intro + Basics.ipynb`** in Google Colab

This part of the work does not require any additional data sets

Github:

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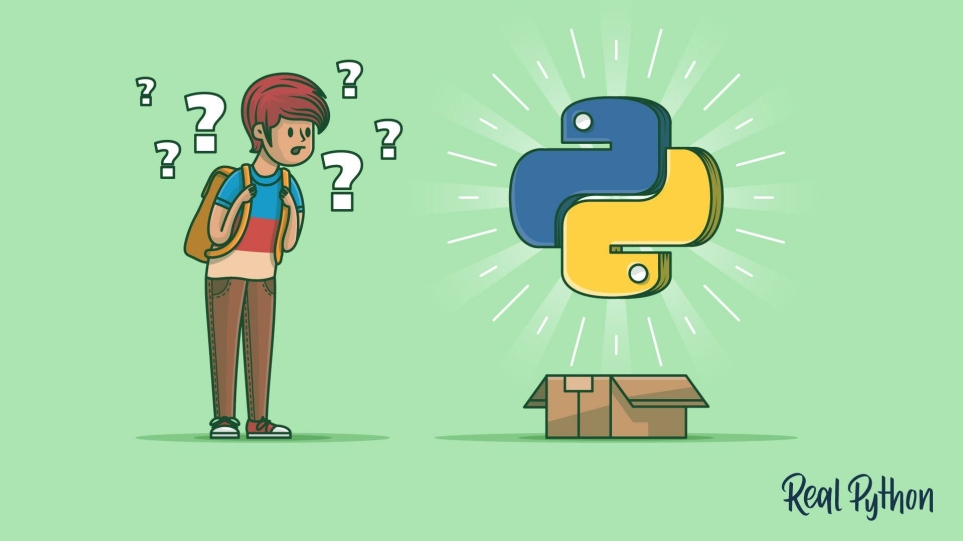

Data manipulation

Open the notebook named **`BMG Workshop – Data manipulation.ipynb`** in Google Colab

To run the code in this notebook, please upload the following data sets into Google Colab:

- **`RacingGameData.csv`**

Github:

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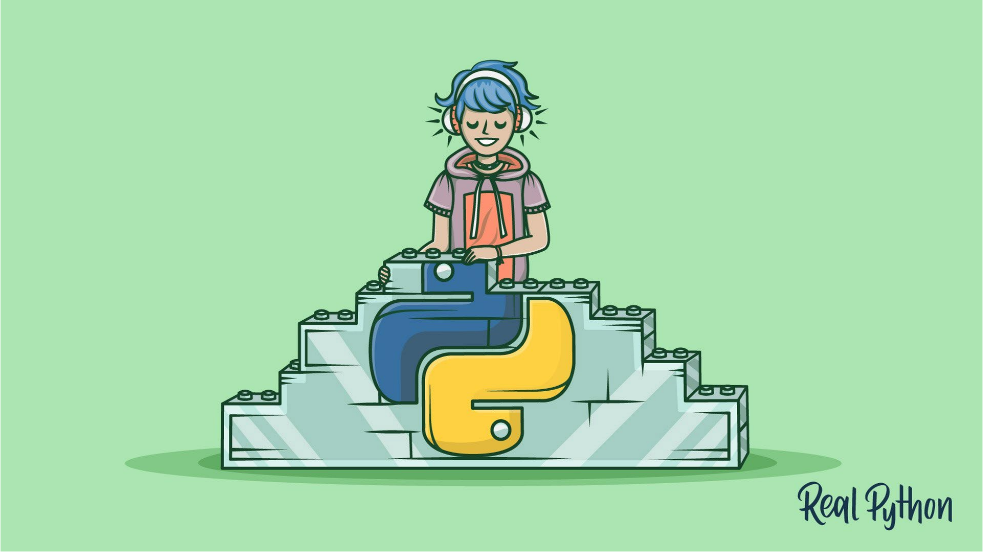

Data visualisation

Open the notebook named **`BMG Workshop – Data visualisation.ipynb`** in Google Colab

To run the code in this notebook, please upload the following data sets into Google Colab:

- **`SuperAnimals.csv`**
- **`Spiders.csv`**

Github:

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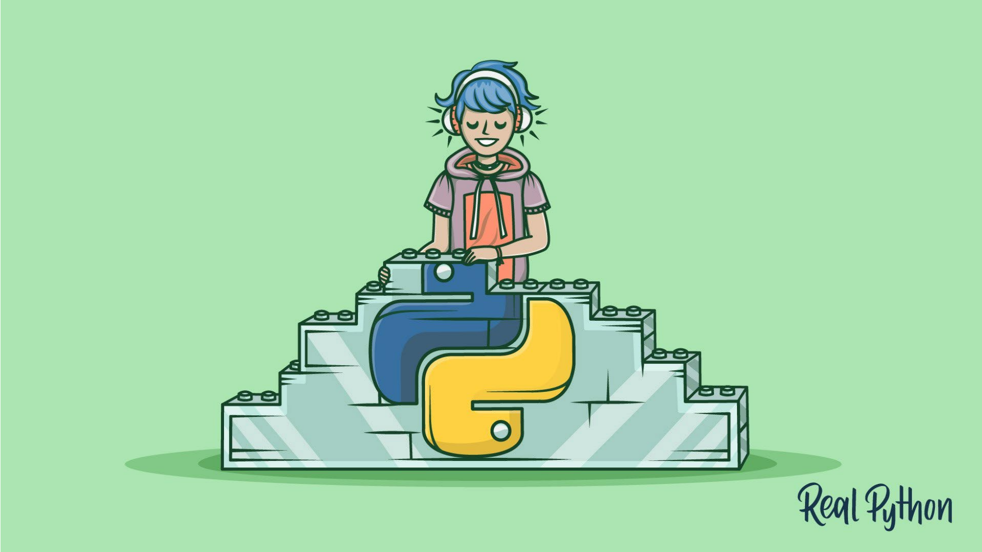

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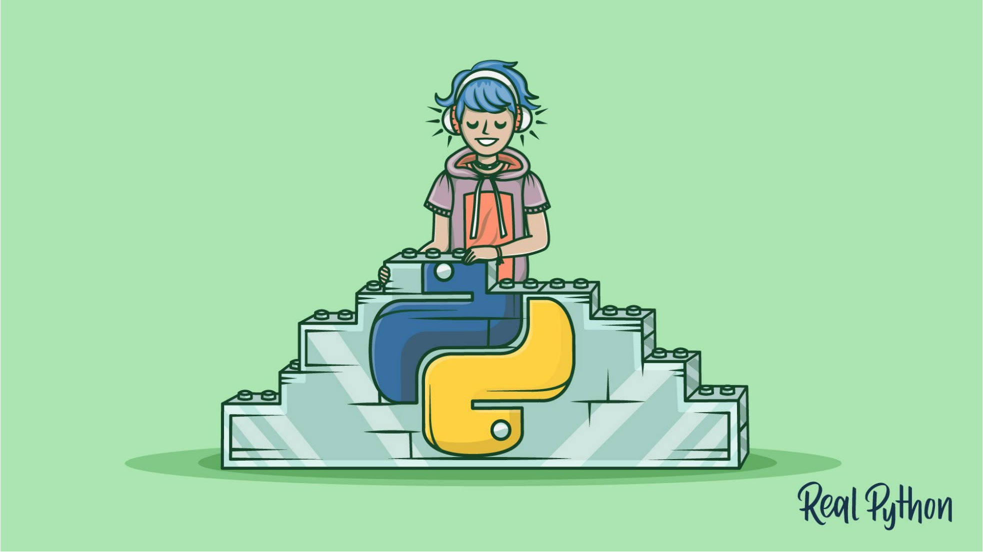

Statistical analysis – Categorical variables

Open the notebook named **`BMG Workshop – Basic statistical analysis – Categorical variables.ipynb`** in Google Colab

This part of the work does not require any additional data sets

Github:

https://github.com/ReneStander/BMG_R_Python_2026

Statistical analysis – Categorical variables

Open the notebook named **`BMG Workshop – Basic statistical analysis – Numerical variables.ipynb`** in Google Colab

To run the code in this notebook, please upload the following data sets into Google Colab:

- **`SuperAnimals.csv`**
- **`Spiders.csv`**

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